Number Theory and Cryptography

4

Chapter Motivation

- *Number theory* is the part of mathematics devoted to the study of the integers and their properties.
- Key ideas in number theory include divisibility and the primality of integers.
- Representations of integers, including binary and hexadecimal representations, are part of number theory.
- Number theory has long been studied because of the beauty of its ideas, its accessibility, and its wealth of open questions.
- We'll use many ideas developed in Chapter 1 about proof methods and proof strategy in our exploration of number theory.
- Mathematicians have long considered number theory to be pure mathematics, but it has important applications to computer science and cryptography studied in Sections 4.5 and 4.6.

Chapter Summary

- Divisibility and Modular Arithmetic
- Integer Representations and Algorithms
- Primes and Greatest Common Divisors
- Solving Congruences
- Applications of Congruences
- Cryptography

4.1 Divisibility and Modular Arithmetic

- Division
- Division Algorithm
- Modular Arithmetic

Division

- $a|b$: If a and b are integers with $a \neq 0$, a divides b (or $a|b$) if there is an integer c such that $b = ac$. If $a|b$, we say that "a is a factor of b" and that "b is a multiple of a".

- Theorem: If a, b, and c are integers:
  - if $a|b$ and $a|c$, then $a|(b + c)$
  - if $a|b$, then $a|bc$ for all integers c
  - if $a|b$ and $b|c$, then $a|c$

- Corollary: If a, b, and c are integers such that $a|b$ and $a|c$, then $a|mb+nc$ whenever m and n are integers.

The Division Algorithm

- Let a and d be integers with d positive. Then there are unique integers q and r, with $0 \leq r < d$ such that $a = dq + r$
  - d is the *divisor*
  - q is the *quotient*
  - r is the *remainder* (has to be positive)
- Notation: $q = a \text{ div } d$, $r = a \bmod d$
- What are the quotient and remainder when –11 is divided by 3?

$$-11 = 3 \cdot (-4) + 1$$

Quotient = –4 and Remainder = 1

Modular Arithmetic [1]

- If a is an integer and m a positive integer, a mod m is the remainder when a is divided by m.
- If $a = qm + r$ and $0 \leq r < m$, then a mod $m = r$

$17 \bmod 5 = 2$ since $17 = 5 \cdot 3 + 2$

$-133 \bmod 9 = 2$ since $-133 = 9 \cdot (-15) + 2$

$2001 \bmod 101 = 82$ since $2001 = 101 \cdot 19 + 82$

Modular Arithmetic [2]

- Definition: Let a and b be integers and m be a positive integer.

  a is congruent to b modulo m if (a - b) is divisible by m.
- Notation: $a \equiv b \pmod{m}$
- Theorem: $a \equiv b \pmod{m}$ iff ($a \bmod m$)=($b \bmod m$)
- Theorem: Let m be a positive integer.

  $a \equiv b \pmod{m}$ iff there is an integer k

  such that $a = b + km$.
- Is 17 congruent to 5 modulo 6 i.e., is $17 \equiv 5 \pmod{6}$?

  (17 – 5) is divisible by 6.

  So the answer is YES.

Congruences of Sums and Products

Theorem 5: Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$

Proof:

- Because $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, by Theorem 4 there are integers s and t with $b = a + sm$ and $d = c + tm$.
- Therefore,
  - $b + d = (a + sm) + (c + tm) = (a + c) + m(s + t)$ and
  - $b\,d = (a + sm)(c + tm) = ac + m(at + cs + stm)$.
- Hence, $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Example: Because $7 \equiv 2 \pmod 5$ and $11 \equiv 1 \pmod 5$, it follows from Theorem 5 that

$18 = 7 + 11 \equiv 2 + 1 = 3 \pmod 5$

$77 = 7 * 11 \equiv 2 + 1 = 3 \pmod 5$

Computing the mod *m* Function of Products and Sums

- We use the following corollary to Theorem 5 to compute the remainder of the product or sum of two integers when divided by *m* from the remainders when each is divided by *m*.

Corollary: Let *m* be a positive integer and let *a* and *b* be integers. Then

$(a + b) \pmod m = ((a \bmod m) + (b \bmod m)) \bmod m$

and

$ab \bmod m = ((a \bmod m)(b \bmod m)) \bmod m$.

(*proof in text:* $a \equiv (a \bmod m) \pmod m$

$b \equiv (b \bmod m) \pmod m$)

Exercises

- **P.258: 30, 38, 41**

4.2 Integer Representations and Algorithms

- Self-study.
- Base *b* expansion of *n*
- Base conversion
- Addition of integer
- Multiplying integers
- Computing div and mod

- Modular exponentiation （模幂运算）

Modular Exponentiation

- **In cryptography, it is important to be able to find b^n mod m efficiently, where b, n, and m are large integers.**
- **Use the binary expansion of n, $n = (a_{k-1},\ldots,a_1,a_0)_2$, to compute b^n Note that:**

$$b^n = b^{a_{k-1}\cdot 2^{k-1}+\cdots+a_1\cdot 2+a_0} = b^{a_{k-1}\cdot 2^{k-1}} \cdots b^{a_1\cdot 2} \cdot b^{a_0}.$$

- **Therefore, to compute b^n, we need only compute the values of b, b^2, $(b^2)^2 = b^4$, $(b^4)^2 = b^8$, …, b^{2^k} and the multiply the terms b^{2^j} in this list, where $a_j = 1$.**

Example: Compute 3^{11} using this method.

Solution: Note that $11 = (1011)_2$ so that $3^{11} = 3^8\, 3^2\, 3^1 = ((3^2)^2)^2\, 3^2\, 3^1 = (9^2)^2 \cdot 9 \cdot 3 = (81)^2 \cdot 9 \cdot 3 = 6561 \cdot 9 \cdot 3 = 117{,}147.$

continued →

Modular Exponentiation

- **The algorithm successively finds b mod m, b^2 mod m, b^4 mod m, …,$b^{2^{k-1}}$ mod m, and multiplies together the terms b^{2^j} where $a_j = 1$.**

procedure *modular exponentiation*(b: integer, $n = (a_{k-1}a_{k-2}\ldots a_1a_0)_2$, m: positive integers)
$x := 1$
$power := b$ **mod** m
for $i := 0$ to $k - 1$
  if $a_i = 1$ **then** $x := (x \cdot power)$ **mod** m
  $power := (power \cdot power)$ **mod** m
return x {x equals b^n **mod** m }

- ***O*((log *m*)^2 log *n*) bit operations are used to find b^n mod *m*.**

Modular Exponentiation

EXAMPLE 12 Use Algorithm 5 to find 3^{644} **mod** 645.

Solution: Algorithm 5 initially sets $x = 1$ and $power = 3$ **mod** $645 = 3$. In the computation of 3^{644} **mod** 645, this algorithm determines 3^{2^j} **mod** 645 for $j = 1, 2, \ldots, 9$ by successively squaring and reducing modulo 645. If $a_j = 1$ (where a_j is the bit in the jth position in the binary expansion of 644, which is $(1010000100)_2$), it multiplies the current value of x by 3^{2^j} **mod** 645 and reduces the result modulo 645. Here are the steps used:

$i = 0$: Because $a_0 = 0$, we have $x = 1$ and $power = 3^2$ **mod** $645 = 9$ **mod** $645 = 9$;

$i = 1$: Because $a_1 = 0$, we have $x = 1$ and $power = 9^2$ **mod** $645 = 81$ **mod** $645 = 81$;

$i = 2$: Because $a_2 = 1$, we have $x = 1 \cdot 81$ **mod** $645 = 81$ and $power = 81^2$ **mod** $645 = 6561$ **mod** $645 = 111$;

$i = 3$: Because $a_3 = 0$, we have $x = 81$ and $power = 111^2$ **mod** $645 = 12{,}321$ **mod** $645 = 66$;

$i = 4$: Because $a_4 = 0$, we have $x = 81$ and $power = 66^2$ **mod** $645 = 4356$ **mod** $645 = 486$;

$i = 5$: Because $a_5 = 0$, we have $x = 81$ and $power = 486^2$ **mod** $645 = 236{,}196$ **mod** $645 = 126$;

$i = 6$: Because $a_6 = 0$, we have $x = 81$ and $power = 126^2$ **mod** $645 = 15{,}876$ **mod** $645 = 396$;

$i = 7$: Because $a_7 = 1$, we find that $x = (81 \cdot 396)$ **mod** $645 = 471$ and $power = 396^2$ **mod** $645 = 156{,}816$ **mod** $645 = 81$;

$i = 8$: Because $a_8 = 0$, we have $x = 471$ and $power = 81^2$ **mod** $645 = 6561$ **mod** $645 = 111$;

$i = 9$: Because $a_9 = 1$, we find that $x = (471 \cdot 111)$ **mod** $645 = 36$.

Exercises

- **P.269: 24, 32, 64**

4.3 Primes and Greatest Common Division

- Prime Numbers and their Properties
- Conjectures and Open Problems About Primes
- Greatest Common Divisors and Least Common Multiples
- The Euclidian Algorithm
- gcds as Linear Combinations

Primes

- A positive integer *p* is called *prime* if the only positive factors of *p* are 1 and *p*. Otherwise *p* is called a *composite*.
- Is 7 prime?
  - The only positive factors are 1 and 7
  - So the answer is YES.
- Is 9 prime?
  - The positive factors are 1, 3, and 9
  - So the answer is NO.

Prime factorization

- The fundamental theorem of arithmetic: Every positive integer can be written uniquely as the product of primes, with the prime factors written in increasing order.

  $100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2^2 \cdot 5^2$

  $641 = 641$

  $999 = 3 \cdot 3 \cdot 3 \cdot 37 = 3^3 \cdot 37$

  $1024 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^{10}$

- Theorem: If n is a composite integer, then n has a prime factor less than or equal to $\sqrt{n}$. (proof ?)
- Show that 101 is prime.
  - **$\sqrt{101} = 10.05$**
  - **The prime numbers less than 10.05 are 2, 3, 5, and 7**
  - **Since 101 is not divisible by any of these, it follows that 101 is prime.**

Procedure for Finding Prime Factorization

- Example: Find the prime factorization of 7007.

  Solution:

  $$7007 = 7 \times 7 \times 11 \times 13$$

The Sieve of Erastosthenes

- **The *Sieve of Erastosthenes* can be used to find all primes not exceeding a specified positive integer. For example, begin with the list of integers between 1 and 100.**
  a. **Delete all the integers, other than 2, divisible by 2.**
  b. **Delete all the integers, other than 3, divisible by 3.**
  c. **Next, delete all the integers, other than 5, divisible by 5.**
  d. **Next, delete all the integers, other than 7, divisible by 7.**
  e. **Since all the remaining integers are not divisible by any of the previous integers, other than 1, the primes are:**

{2,3,5,7,11,13,19,23,29,31,37,41,43,47,53,59,61,67,71,73,79,83 ,89, 97}

The Sieve of Erastosthenes

TABLE 1 The Sieve of Eratosthenes.

Integers divisible by 2 other than 2 receive an underline.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Integers divisible by 3 other than 3 receive an underline.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Integers divisible by 5 other than 5 receive an underline.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Integers divisible by 7 other than 7 receive an underline; integers in color are prime.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

If an integer n is a composite integer, then it has a prime divisor less than or equal to $\sqrt{n}$.

To see this, note that if $n = ab$, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Trial division, a very inefficient method of determining if a number n is prime, is to try every integer $i \leq \sqrt{n}$ and see if n is divisible by i.

The infinitude of primes

- Theorem: There are infinitely many primes.

Proof: Assume finitely many primes: $p_1, p_2, \ldots, p_n$

- **Let $q = p_1 p_2 \cdots p_n + 1$**
- **Either q is prime or by the fundamental theorem of arithmetic it is a product of primes.**
  - **But none of the primes p_j divides q since if $p_j \mid q$, then p_j divides $q - p_1 p_2 \cdots p_n = 1$.**
  - **Hence, there is a prime not on the list $p_1, p_2, \ldots, p_n$. It is either q, or if q is composite, it is a prime factor of q. This contradicts the assumption that $p_1, p_2, \ldots, p_n$ are all the primes.**
- **Consequently, there are infinitely many primes.**

Mersenne Primes

- Mersenne primes（梅森素数）: for almost all the last 300 years, the largest prime known has been an integer of the special form 2^p-1, where *p* is also prime.
- As of 2025, 52 Mersenne primes were known, the largest is $\mathbf{2^{136279841} - 1}$, which has **41,024,320** decimal digits.

Distribution of Primes

- Mathematicians have been interested in the distribution of prime numbers among the positive integers. In the nineteenth century, the *prime number theorem* was proved which gives an asymptotic estimate for the number of primes not exceeding x.

  Prime Number Theorem: The ratio of the number of primes not exceeding x and $x/\ln x$ approaches 1 as x grows without bound. ($\ln x$ is the natural logarithm of x)

  - The theorem tells us that the number of primes not exceeding x, can be approximated by $x/\ln x$.
  - The odds that a randomly selected positive integer less than n is prime are approximately $(n/\ln n)/n = 1/\ln n$.

Primes and Arithmetic Progressions

- Euclid's proof that there are infinitely many primes can be easily adapted to show that there are infinitely many primes in the following $4k + 3$, $k = 1,2,\ldots$ (See Exercise 55)
- In the 19th century G. Lejuenne Dirchlet showed that every arithmetic progression $ka + b$, $k = 1,2, \ldots$, where a and b have no common factor greater than 1 contains infinitely many primes. (The proof is beyond the scope of the text.)
- Are there long arithmetic progressions made up entirely of primes?
  - 5,11, 17, 23, 29 is an arithmetic progression of five primes.
  - 199, 409, 619, 829, 1039,1249,1459,1669,1879,2089 is an arithmetic progression of ten primes.
- In the 1930s, Paul Erdős conjectured that for every positive integer n greater than 1, there is an arithmetic progression of length n made up entirely of primes. This was proven in 2006, by Ben Green and Terrence Tau.（**素数集合中存在任意长度的等差数列**）

Generating Primes

- The problem of generating large primes is of both theoretical and practical interest.
- We will see (in Section 4.6) that finding large primes with hundreds of digits is important in cryptography.
- So far, no useful closed formula that always produces primes has been found. There is no simple function $f(n)$ such that $f(n)$ is prime for all positive integers n.
- But $f(n) = n^2 - n + 41$ is prime for all integers 1,2,…, 40. Because of this, we might conjecture that $f(n)$ is prime for all positive integers n. But $f(41) = 41^2$ is not prime.
- More generally, there is no polynomial with integer coefficients such that $f(n)$ is prime for all positive integers n. (See supplementary Exercise 23.)
- Fortunately, we can generate large integers which are almost certainly primes. See Chapter 7.

Greatest common divisors

- Let *a* and *b* be integers, not both zero. The *greatest common divisor* (gcd) of *a* and *b* is the largest integer *d* such that $d|a$ and $d|b$.
- Notation: $\gcd(a,b) = d$

- Example: What is the gcd of 45 and 60?
  - The positive common divisors are: 1, 3, 5, 15
  - Therefore, $\gcd(45,60) = 15$

Greatest common divisors

- gcd(a,b) can be computed using the prime factorizations of a and b.

$$a = p_1^{a_1} p_2^{a_2} \dots p_n^{a_n}$$
$$b = p_1^{b_1} p_2^{b_2} \dots p_n^{b_n}$$
$$\gcd(a,b) = p_1^{\min(a_1,b_1)} p_2^{\min(a_2,b_2)} \dots p_n^{\min(a_n,b_n)}$$

- Find gcd(120,500).

$$120 = 2^3 \cdot 3 \cdot 5 = 2^3 \cdot 3^1 \cdot 5^1$$
$$500 = 2^2 \cdot 5^3 \quad = 2^2 \cdot 3^0 \cdot 5^3$$
$$\gcd(120,500) = 2^{\min(3,2)} \cdot 3^{\min(1,0)} \cdot 5^{\min(1,3)}$$
$$= 2^2 \cdot 3^0 \cdot 5^1$$
$$= 20$$

Greatest common divisors

- The integers *a* and *b* are *relatively prime* if their greatest common divisor is 1 i.e., gcd(*a*, *b*) = 1.
- Are 17 and 22 are relatively prime?
  - Yes, The reason is: gcd(17,22)=1

- A set of integers a_1, a_2, …, a_n are *pairwise relatively prime* if the gcd of every possible pair is 1.
- Are 10, 17, 21 pairwise relatively prime?
  - Yes, gcd(10,17) = 1, gcd(17,21) = 1, gcd(21,10) = 1
- Are 10, 19, 24 pairwise relatively prime?
  - No, gcd(10,24) = 2

Least Common Multiples

- The *least common multiple* (lcm) of the positive integers a and b is the smallest positive integer m such that $a|m$ and $b|m$.
- Notation: $\text{lcm}(a,b) = m$

- What is the lcm of 6 and 15?

  The common multiples are 30, 60, 90 …..

  Therefore, $\text{lcm}(6,15) = 30$

- Find lcm(120,500).

  $120 = 2^3 \cdot 3 \cdot 5 = 2^3 \cdot 3^1 \cdot 5^1$

  $500 = 2^2 \cdot 5^3 \quad = 2^2 \cdot 3^0 \cdot 5^3$

  $\text{lcm}(120,500) = 2^{\max(3,2)} \cdot 3^{\max(1,0)} \cdot 5^{\max(1,3)}$

  $= 2^3 \cdot 3^1 \cdot 5^3 = 8 \cdot 3 \cdot 125 = 3000$

Relationship between gcd and lcm

- Theorem: If a and b are positive integers, then

$$ab = \gcd(a,b) \cdot \mathrm{lcm}(a,b)$$

- Example

$$\gcd(120,500) \cdot \mathrm{lcm}(120,500)$$
$$= 20 * 3000$$
$$= 60000$$
$$= 120 \cdot 500$$

The Euclidean algorithm

Procedure gcd(*a*,*b*: positive integers)

x:=*a*

y:=*b*

while *y*≠0

begin

    r:=*x* mod *y*

    x:=*y*

    y:=*r*

end {gcd(*a*,*b*) is *x*}

gcd(414, 662)

662 = 414 • 1 + 248

414 = 248 • 1 + 166

248 = 166 • 1 + 82

166 = 82 • 2 + 2

82 = 2 • 41

gcd(414, 662)=2

Correctness of Euclidean Algorithm

Lemma 1: Let $a = bq + r$, where a, b, q, and r are integers. Then $\gcd(a,b) = \gcd(b,r)$.

Proof:

- **Suppose that d divides both a and b. Then d also divides $a - bq = r$ (by Theorem 1 of Section 4.1). Hence, any common divisor of a and b must also be any common divisor of b and r.**
- **Suppose that d divides both b and r. Then d also divides $bq + r = a$. Hence, any common divisor of b and r must also be a common divisor of a and b.**
- **Therefore, $\gcd(a,b) = \gcd(b,r)$.**

Correctness of Euclidean Algorithm

- **Suppose that a and b are positive integers with $a \geq b$.**
  Let $r_0 = a$ and $r_1 = b$.
  Successive applications of the division algorithm yields:

$$
\begin{aligned}
r_0 &= r_1 q_1 + r_2 && 0 \leq r_2 < r_1, \\
r_1 &= r_2 q_2 + r_3 && 0 \leq r_3 < r_2, \\
&\cdots \\
r_{n-2} &= r_{n-1} q_{n-1} + r_2 && 0 \leq r_n < r_{n-1}, \\
r_{n-1} &= r_n q_n . &&
\end{aligned}
$$

- **Eventually, a remainder of zero occurs in the sequence of terms: $a = r_0 > r_1 > r_2 > \cdots \geq 0$. The sequence can't contain more than a terms.**
- **By Lemma 1**

  $$\gcd(a,b) = \gcd(r_0,r_1) = \cdots = \gcd(r_{n-1},r_n) = \gcd(r_n, 0) = r_n.$$
- **Hence the greatest common divisor is the last nonzero remainder in the sequence of divisions.**

gcds as Linear Combinations

Bézout's Theorem: If *a* and *b* are positive integers, then there exist integers *s* and *t* such that gcd(*a*,*b*) = *sa* + *tb*.

(*proof in exercises of Section* 5.2)

Definition: If *a* and *b* are positive integers, then integers *s* and *t* such that gcd(*a*,*b*) = *sa* + *tb* are called *Bézout coefficients* of *a* and *b*. The equation gcd(*a*,*b*) = *sa* + *tb* is called *Bézout's identity*.

- **By Bézout's Theorem, the gcd of integers *a* and *b* can be expressed in the form *sa* + *tb* where *s* and *t* are integers. This is a *linear combination* with integer coefficients of *a* and *b*.**
  - **gcd(6,14) = (−2)·6 + 1·14**

Finding gcds as Linear Combinations

Example: Express gcd(252,198) = 18 as a linear combination of 252 and 198.

Solution: First use the Euclidean algorithm to show gcd(252,198) = 18

i. $252 = 1 \cdot 198 + 54$

ii. $198 = 3 \cdot 54 + 36$

iii. $54 = 1 \cdot 36 + 18$

iv. $36 = 2 \cdot 18$

- Now working backwards, from iii and i above
  - $18 = 54 - 1 \cdot 36$
  - $36 = 198 - 3 \cdot 54$
- Substituting the 2nd equation into the 1st yields:
  - $18 = 54 - 1 \cdot (198 - 3 \cdot 54) = 4 \cdot 54 - 1 \cdot 198$
- Substituting $54 = 252 - 1 \cdot 198$ (from i)) yields:
  - $18 = 4 \cdot (252 - 1 \cdot 198) - 1 \cdot 198 = 4 \cdot 252 - 5 \cdot 198$

Exercises

- **P.289: 17, 28, 32, 42, 55**

4.4 Solving Congruences

- $$ax \equiv b \pmod{m}$$

Linear Congruences

Definition: A congruence of the form

$$ax \equiv b \pmod{m},$$

where m is a positive integer, a and b are integers, and x is a variable, is called a *linear congruence*.

- **The solutions to a linear congruence $ax \equiv b \pmod{m}$ are all integers x that satisfy the congruence.**

Definition: An integer $\bar{a}$ such that $\bar{a}a \equiv 1 \pmod{m}$ is said to be an *inverse* of a modulo m.

Example: 5 is an inverse of 3 modulo 7 since $5 \cdot 3 = 15 \equiv 1 \pmod{7}$

- **One method of solving linear congruences makes use of an inverse $\bar{a}$, if it exists. Although we can not divide both sides of the congruence by a, we can multiply by $\bar{a}$ to solve for x.**

Inverse of *a* modulo *m*

- **The following theorem guarantees that an inverse of *a* modulo *m* exists whenever *a* and *m* are relatively prime. Two integers *a* and *b* are relatively prime when gcd(*a*,*b*) = 1.**

Theorem 1: If *a* and *m* are relatively prime integers and *m* > 1, then an inverse of *a* modulo *m* exists. Furthermore, this inverse is unique modulo *m*. (This means that there is a unique positive integer $\bar{a}$ less than *m* that is an inverse of *a* modulo *m* and every other inverse of *a* modulo *m* is congruent to $\bar{a}$ modulo *m*.)

Proof: Since gcd(*a*,*m*) = 1, by Theorem 6 of Section 4.3, there are integers *s* and *t* such that $sa + tm = 1$.

- **Hence, $sa + tm \equiv 1 \pmod{m}$.**
- **Since $tm \equiv 0 \pmod{m}$, it follows that $sa \equiv 1 \pmod{m}$**
- **Consequently, *s* is an inverse of *a* modulo *m*.**
- **The uniqueness of the inverse is Exercise 7.**

Finding Inverses

- **The Euclidean algorithm and Bézout coefficients gives us a systematic approaches to finding inverses.**

  Example: Find an inverse of 3 modulo 7.

  Solution: Because gcd(3,7) = 1, by Theorem 1, an inverse of 3 modulo 7 exists.

  - **Using the Euclidian algorithm: $7 = 2 \cdot 3 + 1$.**
  - **From this equation, we get $-2 \cdot 3 + 1 \cdot 7 = 1$, and see that -2 and 1 are Bézout coefficients of 3 and 7.**
  - **Hence, -2 is an inverse of 3 modulo 7.**
  - **Also every integer congruent to -2 modulo 7 is an inverse of 3 modulo 7, i.e., 5, -9, 12, etc.**

Finding Inverses

Example: Find an inverse of 101 modulo 4620.

Solution: First use the Euclidian algorithm to show that gcd(101,4620) = 1.

$4620 = 45 \cdot 101 + 75$
$101 = 1 \cdot 75 + 26$
$75 = 2 \cdot 26 + 23$
$26 = 1 \cdot 23 + 3$
$23 = 7 \cdot 3 + 2$
$3 = 1 \cdot 2 + 1$
$2 = 2 \cdot 1$

Working Backwards:

$1 = 3 - 1 \cdot 2$
$1 = 3 - 1 \cdot (23 - 7 \cdot 3) = -1 \cdot 23 + 8 \cdot 3$
$1 = -1 \cdot 23 + 8 \cdot (26 - 1 \cdot 23) = 8 \cdot 26 - 9 \cdot 23$
$1 = 8 \cdot 26 - 9 \cdot (75 - 2 \cdot 26) = 26 \cdot 26 - 9 \cdot 75$
$1 = 26 \cdot (101 - 1 \cdot 75) - 9 \cdot 75$
$= 26 \cdot 101 - 35 \cdot 75$
$1 = 26 \cdot 101 - 35 \cdot (4620 - 45 \cdot 101)$
$= -35 \cdot 4620 + 1601 \cdot 101$

Since the last nonzero remainder is 1, gcd(101,4620) = 1

Bézout coefficients : -35 and 1601

1601 is an inverse of 101 modulo 4620

Using Inverses to Solve Congruences

- **We can solve the congruence $ax \equiv b \pmod{m}$ by multiplying both sides by $\bar{a}$.**

Example: What are the solutions of the congruence $3x \equiv 4 \pmod{7}$.

Solution: We found that -2 is an inverse of 3 modulo 7. We multiply both sides of the congruence by -2 giving

$$-2 \cdot 3x \equiv -2 \cdot 4 \pmod{7}.$$

Because $-6 \equiv 1 \pmod{7}$ and $-8 \equiv 6 \pmod{7}$, it follows that if x is a solution, then $x \equiv -8 \equiv 6 \pmod{7}$

We need to determine if every x with $x \equiv 6 \pmod{7}$ is a solution. Assume that $x \equiv 6 \pmod{7}$. By Theorem 5 of Section 4.1, it follows that $3x \equiv 3 \cdot 6 = 18 \equiv 4 \pmod{7}$ which shows that all such x satisfy the congruence.

The solutions are the integers x such that $x \equiv 6 \pmod{7}$, namely, 6,13,20 … and $-1, -8, -15, \ldots$

The Chinese Remainder Theorem

Theorem 2: (*The Chinese Remainder Theorem*) Let $m_1, m_2, \ldots, m_n$ be pairwise relatively prime positive integers greater than one and $a_1, a_2, \ldots, a_n$ arbitrary integers. Then the system

$$x \equiv a_1 \pmod{m_1}$$
$$x \equiv a_2 \pmod{m_2}$$
$$\vdots$$
$$x \equiv a_n \pmod{m_n}$$

has a unique solution modulo $m = m_1 m_2 \cdots m_n$.

(That is, there is a solution x with $0 \leq x < m$ and all other solutions are congruent modulo m to this solution.)

中国余数问题

最早记载于南北朝时期的数学著作《孙子算经》中的 “物不知数” 问题：

原文：今有物不知其数，三三数之剩二，五五数之剩三，七七数之剩二。问物几何?

译文：一个数，除以 3 余 2，除以 5 余 3，除以 7 余 2，求这个数。

解法口诀（明朝程大位《算法统宗》）：

三人同行七十稀，五树梅花廿一枝，七子团圆正半月，除百零五便得知。

1. 三人同行七十稀

三人同行：对应原题中 **除以 3 取余数** 的条件（x≡2(mod3)）。

七十稀：指数字 **70**，这个数的特点是：

能被 5 和 7 整除（是 5、7 的最小公倍数 35 的 2 倍）；

除以 3 余 1（70÷3=23…1）。

作用：用原题中 “除以 3 的余数”（即 2）乘以 70，得到140。

2. 五树梅花廿一枝

五树梅花：对应原题中 **除以 5 取余数** 的条件（x≡3(mod5)）。

廿一枝：指数字 **21**，这个数的特点是：

能被 3 和 7 整除（是 3、7 的最小公倍数 21）；

除以 5 余 1（21÷5=4…1）。

作用：用原题中 “除以 5 的余数”（即 3）乘以 21，得到63。

3. 七子团圆正半月

七子团圆：对应原题中 **除以 7 取余数** 的条件（x≡2(mod7)）。

正半月：“半月” 指 15 天，对应数字 **15**，

作用：用原题中 “除以 7 的余数”（即 2）乘以 15，得到30。

4. 除百零五便得知

百零五：指数字 **105**，是 3、5、7 的最小公倍数（3×5×7=105）

除：取模运算（即除以 105 取余数）。

作用：把前 3 步得到的结果相加，再对 105 取余，就是最小正整数解：(140+63+30)(mod105)=233(mod105)=23

口诀本质是构造出 **“对一个模数余 1，对其他模数余 0”** 的基准数（70、21、15），再用各自的余数加权求和，最后通过最小公倍数取模得到唯一解。

The Chinese Remainder Theorem

To construct a solution first let $M_k = m/m_k$ for $k = 1,2,\ldots,n$ and $m = m_1 m_2 \cdots m_n$.

Since $\gcd(m_k, M_k) = 1$, by Theorem 1, there is an integer y_k, an inverse of M_k modulo m_k, such that

$$M_k y_k \equiv 1 \pmod{m_k}.$$

Form the sum

$$x = a_1 M_1 y_1 + a_2 M_2 y_2 + \cdots + a_n M_n y_n .$$

Note that because $M_j \equiv 0 \pmod{m_k}$ whenever $j \neq k$, all terms except the kth term in this sum are congruent to 0 modulo m_k.

Because $M_k y_k \equiv 1 \pmod{m_k}$, we see that $x \equiv a_k M_k y_k \equiv a_k \pmod{m_k}$, for $k = 1,2,\ldots,n$.

Hence, x is a simultaneous solution to the n congruences.

$$x \equiv a_1 \pmod{m_1} \quad x \equiv a_2 \pmod{m_2} \quad \ldots \quad x \equiv a_n \pmod{m_n}$$

The Chinese Remainder Theorem

Example: Consider the 3 congruences from Sun-Tsu's problem:

$x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 5$, $x \equiv 2 \pmod 7$.

- Let $m = 3 \cdot 5 \cdot 7 = 105$, $M_1 = m/3 = 35$, $M_3 = m/5 = 21$, $M_3 = m/7 = 15$.
- We see that
  - 2 is an inverse of $M_1 = 35$ modulo 3 since $35 \cdot 2 \equiv 2 \cdot 2 \equiv 1 \pmod 3$
  - 1 is an inverse of $M_2 = 21$ modulo 5 since $21 \equiv 1 \pmod 5$
  - 1 is an inverse of $M_3 = 15$ modulo 7 since $15 \equiv 1 \pmod 7$
- Hence,

$$x = a_1M_1y_1 + a_2M_2y_2 + a_3M_3y_3$$
$$= 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 = 233 \equiv 23 \pmod{105}$$

Fermat's Little Theorem

Theorem 3: (*Fermat's Little Theorem*) If p is prime and a is an integer not divisible by p, then $a^{p-1} \equiv 1 \pmod{p}$

Furthermore, for every integer a we have $a^p \equiv a \pmod{p}$

(*proof outlined in Exercise 19*)

Fermat's little theorem is useful in computing the remainders modulo p of large powers of integers.

Example: Find 7^{222} mod 11.

By Fermat's little theorem, we know that $7^{10} \equiv 1 \pmod{11}$, and so $(7^{10})^k \equiv 1 \pmod{11}$, for every positive integer k. Therefore,

$$7^{222} = 7^{22 \cdot 10 + 2} = (7^{10})^{22} 7^2 \equiv (1)^{22} \cdot 49 \equiv 5 \pmod{11}.$$

Hence, 7^{222} mod 11 = 5.

Exercises

- **P.300: 12, 14, 19**

4.5 Applications of Congruences

Self-study

No in the scope of examination

Hashing Functions

Definition: A *hashing function h* assigns memory location *h*(*k*) to the record that has *k* as its key.

- **A common hashing function is *h*(*k*) = *k* mod *m*, where *m* is the number of memory locations.**
- **Because this hashing function is onto, all memory locations are possible.**

Example: Let *h*(*k*) = *k* mod 111. This hashing function assigns the records of customers with social security numbers as keys to memory locations in the following manner:

h(064212848) = 064212848 mod 111 = 14

h(037149212) = 037149212 mod 111 = 65

h(107405723) = 107405723 mod 111 = 14, but since location 14 is already occupied, the record is assigned to the next available position, which is 15.

- **The hashing function is not one-to-one as there are many more possible keys than memory locations. When more than one record is assigned to the same location, we say a *collision* occurs. Here a collision has been resolved by assigning the record to the first free location.**

Pseudorandom Numbers

- **Randomly chosen numbers are needed for many purposes, including computer simulations.**
- ***Pseudorandom numbers* are not truly random since they are generated by systematic methods.**
- **The *linear congruential method* is one commonly used procedure for generating pseudorandom numbers.**
- **Four integers are needed: the *modulus m*, the *multiplier a*, the *increment c*, and *seed* x_0, with $2 \leq a < m$, $0 \leq c < m$, $0 \leq x_0 < m$.**
- **We generate a sequence of pseudorandom numbers $\{x_n\}$, with $0 \leq x_n < m$ for all n, by successively using the recursively defined function**

$$x_{n+1} = (ax_n + c) \bmod m.$$

(*an example of a recursive definition, discussed in Section* 5.3*)*

- **If psudorandom numbers between 0 and 1 are needed, then the generated numbers are divided by the modulus, x_n/m.**

Pseudorandom Numbers

- **Example: Find the sequence of pseudorandom numbers generated by the linear congruential method with modulus $m = 9$, multiplier $a = 7$, increment $c = 4$, and seed $x_0 = 3$.**
- **Solution: Compute the terms of the sequence by successively using the congruence $x_{n+1} = (7x_n + 4) \bmod 9$, with $x_0 = 3$.**

$$x_1 = 7x_0 + 4 \bmod 9 = 7\cdot 3 + 4 \bmod 9 = 25 \bmod 9 = 7,$$
$$x_2 = 7x_1 + 4 \bmod 9 = 7\cdot 7 + 4 \bmod 9 = 53 \bmod 9 = 8,$$
$$x_3 = 7x_2 + 4 \bmod 9 = 7\cdot 8 + 4 \bmod 9 = 60 \bmod 9 = 6,$$
$$x_4 = 7x_3 + 4 \bmod 9 = 7\cdot 6 + 4 \bmod 9 = 46 \bmod 9 = 1,$$
$$x_5 = 7x_4 + 4 \bmod 9 = 7\cdot 1 + 4 \bmod 9 = 11 \bmod 9 = 2,$$
$$x_6 = 7x_5 + 4 \bmod 9 = 7\cdot 2 + 4 \bmod 9 = 18 \bmod 9 = 0,$$
$$x_7 = 7x_6 + 4 \bmod 9 = 7\cdot 0 + 4 \bmod 9 = 4 \bmod 9 = 4,$$
$$x_8 = 7x_7 + 4 \bmod 9 = 7\cdot 4 + 4 \bmod 9 = 32 \bmod 9 = 5,$$
$$x_9 = 7x_8 + 4 \bmod 9 = 7\cdot 5 + 4 \bmod 9 = 39 \bmod 9 = 3.$$

The sequence generated is 3,7,8,6,1,2,0,4,5,3,7,8,6,1,2,0,4,5,3,...

Check Digits: UPCs

- **A common method of detecting errors in strings of digits is to add an extra digit at the end, which is evaluated using a function. If the final digit is not correct, then the string is assumed not to be correct.**

Example: Retail products are identified by their *Universal Product Codes* (*UPC*s). Usually these have 12 decimal digits, the last one being the check digit. The check digit is determined by the congruence:

$$3x_1 + x_2 + 3x_3 + x_4 + 3x_5 + x_6 + 3x_7 + x_8 + 3x_9 + x_{10} + 3x_{11} + x_{12} \equiv 0 \pmod{10}.$$

a. Suppose that the first 11 digits of the UPC are 79357343104. What is the check digit?

Solution:

a. $3\cdot 7 + 9 + 3\cdot 3 + 5 + 3\cdot 7 + 3 + 3\cdot 4 + 3 + 3\cdot 1 + 0 + 3\cdot 4 + x_{12} \equiv 0 \pmod{10}$

$21 + 9 + 9 + 5 + 21 + 3 + 12 + 3 + 3 + 0 + 12 + x_{12} \equiv 0 \pmod{10}$

$98 + x_{12} \equiv 0 \pmod{10}$

$x_{12} \equiv 0 \pmod{10}$ So, the check digit is 2.

Exercises

- **P.292: 1, 18**

4.6 Cryptology

- Encryption and decryption

Caesar cipher

- a→0,b→1,…,z →25
- *Encryption function:* f(p) = (p+3) mod 26
- *Decryption function:* f(p) = (p-3) mod 26
- Plaintext: meet you in the park at six
- Solution: replace letters with numbers:
- 12 4 4 19 24 14 20 8 13 19 7 4
  15 0 17 10 0 19 18 8 23 15 7 7 22 27 17 23 11 16 22 10 7
  18 3 20 13 3 22 21 11 1

  Ciphertext: phhw brx lq wkh sdun dw vlb

Caesar cipher

- The Caesar cipher is one of a family of ciphers called *shift ciphers.* Letters can be shifted by an integer *k,* with 3 being just one possibility. The encryption function is

$$f(p) = (p + k) \bmod 26$$

and the decryption function is

$$f^{-1}(p) = (p - k) \bmod 26$$

The integer *k* is called a *key*.

Affine Cipher

- Shift ciphers are a special case of *affine ciphers* which use functions of the form $f(p) = (ap + b) \bmod 26$,

  where a and b are integers, chosen so that f is a bijection.

  The function is a bijection if and only if $\gcd(a,26) = 1$. Where a and 26 are relatively prime, we usually take a prime, for example $f(p)=(7p+3) \bmod 26$
- Plaintext: meet you in the park at six

  Encryption function: $f(p) = (7p+3) \bmod 26$

  replace letters with numbers:

  12 4 4 19 24 14 20 8 13 19 7 4

  15 0 17 10 0 19 18 8 23

  9 5 5 6 15 23 13 7 16 10 0 5

  4 3 18 21 3 10 25 7 8

  Ciphertext: jffg pxn hq kaf edtw dk zhi

Affine Cipher

- To decrypt a message encrypted by a shift cipher, the congruence $c \equiv ap + b \pmod{26}$ needs to be solved for p.
  - Subtract b from both sides to obtain $c - b \equiv ap \pmod{26}$.
  - Multiply both sides by the inverse of a modulo 26, which exists since gcd(a,26) = 1.
  - $\bar{a}(c - b) \equiv \bar{a}ap \pmod{26}$, which simplifies to $\bar{a}(c - b) \equiv p \pmod{26}$.
  - $p \equiv \bar{a}(c - b) \pmod{26}$ is used to determine p in Z_{26}.

Cryptanalysis of Affine Ciphers

- **The process of recovering plaintext from ciphertext without knowledge both of the encryption method and the key is known as *cryptanalysis* or *breaking codes*.**
- **An important tool for cryptanalyzing ciphertext produced with a affine ciphers is the relative frequencies of letters. The nine most common letters in the English texts are E 13%, T 9%, A 8%, O 8%, I 7%, N 7%, S 7%, H 6%, and R 6%.**
- **To analyze ciphertext:**
  - **Find the frequency of the letters in the ciphertext.**
  - **Hypothesize that the most frequent letter is produced by encrypting E.**
  - **If the value of the shift from E to the most frequent letter is k, shift the ciphertext by $-k$ and see if it makes sense.**
  - **If not, try T as a hypothesis and continue.**

Public Key Cryptography

- All classical ciphers, including shift and affine ciphers, are *private key cryptosystems*. Knowing the encryption key allows one to quickly determine the decryption key.
- All parties who wish to communicate using a private key cryptosystem must share the key and keep it a secret.
- In public key cryptosystems, first invented in the 1970s, knowing how to encrypt a message does not help one to decrypt the message. Therefore, everyone can have a publicly known encryption key. The only key that needs to be kept secret is the decryption key.

The RSA Cryptosystem

- A public key cryptosystem, now known as the RSA system was introduced in 1976 by three researchers at MIT. (Ronald Rivest, Adi Shamir, Leonard Adelman)
- It is now known that the method was discovered earlier by Clifford Cocks, working secretly for the UK government.
- The public encryption key is (n,e), where $n = pq$ (the modulus) is the product of two large (200 digits) primes p and q, and an exponent e that is relatively prime to $(p-1)(q-1)$. The two large primes can be quickly found using probabilistic primality tests, discussed earlier. But $n = pq$, with approximately 400 digits, cannot be factored in a reasonable length of time.

RSA Encryption

- **To encrypt a message using RSA using a key (n,e) :**
  - i. **Translate the plaintext message M into sequences of two digit integers representing the letters. Use 00 for A, 01 for B, etc.**
  - ii. **Concatenate the two digit integers into strings of digits.**
  - iii. **Divide this string into equally sized blocks of $2N$ digits where $2N$ is the largest even number 2525…25 with $2N$ digits that does not exceed n.**
  - iv. **The plaintext message M is now a sequence of integers $m_1,m_2,\ldots,m_k$.**
  - v. **Each block (an integer) is encrypted using the function $C = M^e \bmod n$.**

Example: Encrypt the message STOP using the RSA cryptosystem with key(2537,13).

- **$2537 = 43 \cdot 59$,**
- **$p = 43$ and $q = 59$ are primes and $\gcd(e,(p-1)(q-1)) = \gcd(13, 42 \cdot 58) = 1$.**

Solution: Translate the letters in STOP to their numerical equivalents 18 19 14 15.

- **Divide into blocks of four digits (because 2525 < 2537 < 252525) to obtain 1819 1415.**
- **Encrypt each block using the mapping $C = M^{13} \bmod 2537$.**
- **Since $1819^{13} \bmod 2537 = 2081$ and $1415^{13} \bmod 2537 = 2182$, the encrypted message is 2081 2182.**

RSA Decryption

- **To decrypt a RSA ciphertext message, the decryption key *d*, an inverse of *e* modulo $(p-1)(q-1)$ is needed. The inverse exists since $\gcd(e,(p-1)(q-1)) = \gcd(13, 42 \cdot 58) = 1$.**
- **With the decryption key *d*, we can decrypt each block with the computation $M = C^d \bmod p \cdot q$. (*see text for full derivation*)**
- **RSA works as a public key system since the only known method of finding *d* is based on a factorization of *n* into primes. There is currently no known feasible method for factoring large numbers into primes.**

Example: The message 0981 0461 is received. What is the decrypted message if it was encrypted using the RSA cipher from the previous example.

Solution: The message was encrypted with $n = 43 \cdot 59$ and exponent 13. An inverse of 13 modulo $42 \cdot 58 = 2436$ (*exercise* 2 *in Section* 4.4) is $d = 937$.

- **To decrypt a block *C*, $M = C^{937} \bmod 2537$.**
- **Since $0981^{937} \bmod 2537 = 0704$ and $0461^{937} \bmod 2537 = 1115$, the decrypted message is 0704 1115. Translating back to English letters, the message is HELP.**

Exercises

- **P.304: 3, 11**